Controlled Testing and Failure Attribution for ML Sensor Pipelines

Two-page Paper 3 proposal

Working proposal for postdoc and advisor review

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Abstract

Post-training monitors can flag distribution change but rarely distinguish a sensor/acquisition shift from unsupported physical states or a broken evaluation contract. We propose a controlled, replay-based study of that distinction—not a maintained software product. Declared interventions, named representation hooks, deterministic contract gates, and calibrated abstention are evaluated across NuBench point clouds and a fixed waveform reconstruction task. Against input tests, MMD, classifier two-sample tests, embedding distance, uncertainty, and output-only monitoring, we measure detection power and delay, false alerts, failure-family attribution, latency, storage, and reproducibility. The intended result is quantitative: generic monitors detect N/S shifts, layerwise evidence improves attribution, and deterministic gates isolate evaluation faults at measured replay cost.

1 Scope, continuity, and novelty

Paper 1 identifies likelihood geometry under structured noise; Paper 2 separates residual geometry, architecture class, and training support. Paper 3 asks whether those contracts can be tested after training. Its artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a branded platform or a promise of production maintenance.

The contribution is a controlled cross-modal design that combines (i) paired sensor perturbations, support-composition shifts, and deliberate contract faults; (ii) feature-matched attribution comparisons with guarded abstention; and (iii) a power–cost study of full capture, sampling, reservoirs, and fixed-dimensional sketches. NuBench, the waveform-noise generator, and MMD are inputs to that design, not claimed inventions. Cross-geometry results test consistency, not causal detector-design effects.

2 Failure contracts and claims

We predeclare three labels. **N—acquisition/noise shift** includes PSD or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, and channel/module loss. **S—training-support shift** changes window composition toward physical strata that are sparse under the clean training distribution; events themselves are not corrupted. **E—evaluation- contract fault** includes wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. E is not a distribution shift and is not the likelihood-metric mismatch studied in Papers 1–2.

C1—Detect N/S shifts. At a 1% false-alert rate on nominal, non-overlapping windows, the best monitoring signature detects at least 80% of held-out abrupt and gradual N/S shifts within five windows and is non-inferior to the strongest eligible baseline by a five-point power margin. Delay is primary in events and simulated seconds.

C2—Attribute with abstention. A fixed classifier over statistical monitors plus deterministic gates reaches macro-F1 ≥ 0.80 on held-out N/S/E cases. For N/S, the full layerwise signature exceeds the same classifier using all generic-monitor outputs by ≥ 0.20 F1. Split-conformal distance in standardized fingerprint space must identify an unseen family with AUROC ≥ 0.80 instead of forcing a known label. This is intervention-family attribution, not unrestricted physical root-cause discovery.

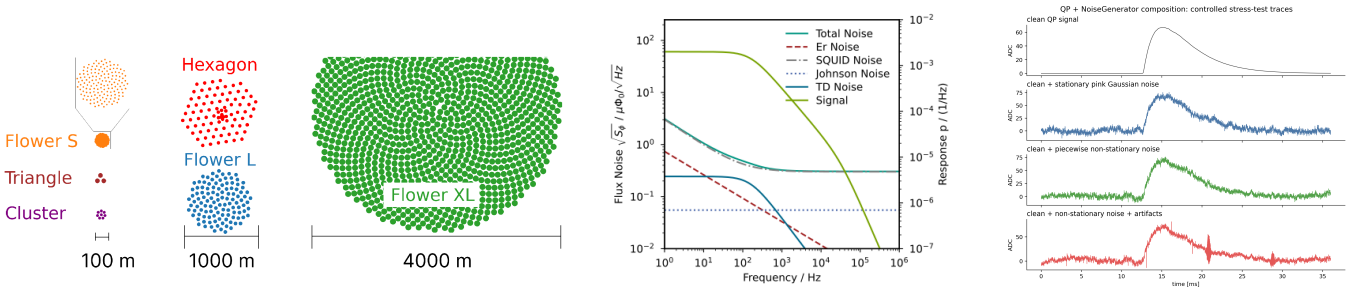
C3—Trade power for cost. Ten-percent event sampling plus fixed-size sketches yields p95 replay-path latency overhead $< 10\%$, at least $10\times$ less representation storage, and no more than five points of power/F1 loss relative to full capture. Effects and intervals are repeated across modalities, three NuBench model families, two machines, and CPU/GPU where supported. These targets are frozen after development profiling and power analysis.

3 Methods and testbeds

Each event window records input-quality summaries, named-layer moments or sketches, outputs, uncertainty, delayed task metrics, and calibration. Eligible N/S baselines receive identical windows and false-alert budgets: corrected univariate input tests, RBF-MMD [2], a classifier two-sample test [3], final-embedding mean/covariance distance, output tests, and uncertainty/entropy. A regularized logistic classifier is fitted with identical splits to five arms: input only; output/uncertainty; final embedding only; all generic-monitor outputs; and the full layerwise signature. E uses schema/range invariants and clean-reference tests, not drift statistics. Clean miscalibration is a model result, not automatically a software bug [4]; latent displacement is normalized by clean within-class dispersion.

NuBench point clouds. The first cell is Hexagon Ice LE $\times$ DynEdge $\times$ direction. The restored 1,358,099-parameter checkpoint exposes a 128-D pre-head event representation. Module dropout, hit thinning, and time jitter instantiate N. S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training only. ParticleNeT and GRIT repeat the study; another geometry is a consistency check [1].

Waveforms and the noise module. The fixed task uses non-overlapping 4096-sample windows at 10 MS/s, a rank-2 noise-aware EMPCA encoder fitted on the chronological training split, and sinusoid-amplitude reconstruction. The



(a) NuBench detector geometries (b) detector signal/noise spectrum (c) generated waveform stress tests

Figure 1: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The supplied Al_2O_3/Al athermal-detector response and component PSDs. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

module accepts or constructs PSD/CSD models, samples seeded frequency-domain random coefficients with the required real-signal symmetry, transforms them into time traces, and adds stationary color, piecewise nonstationarity, lines, drift, glitches, and tails to clean signals (Fig. 1b–c). It validates units and second-order spectral statistics. It is therefore a controlled N intervention engine, not another dataset, reconstruction model, or physical- source separator. S holds out declared frequency/amplitude ranges; E swaps coefficient order or frequency units and changes PSD/score conventions.

Table 1: Confirmatory experiment matrix. Drift baselines apply only to N/S.

ID	Test	Primary endpoints
E0	Clean replay; deliberate E faults	Reference discrepancy; gate sensitivity/specificity
E1	Held-out abrupt/gradual N/S	Power at 1% false alert; AUROC/AUPRC; delay
E2	N/S/E, mixtures, unseen family	Macro/multi-label F1; confusion; unknown AUROC
E3	Capture vs sample/reservoir/sketch	p50/p95 latency; bytes/event; power/F1 loss
E4	Models, modalities, hardware	Scaling; interval agreement; report hashes

Held-out severities, seeds, support strata, N+S mixtures, and one unseen family prevent perturbation lookup. Windows and thresholds are frozen after development. Intervals resample event groups and perturbation seeds separately. Cost sweeps event rate, window size, hook count, embedding dimension, and model count; repeat runs record machine, accelerator, timing method, and data/config hashes. Null results require a passed positive control, adequate precision, and a predeclared equivalence margin.

4 Smoke test and feasibility gate

The selected 8.82 GiB NuBench release is checksummed; the DynEdge checkpoint restores and the 128-D hook runs. Rescoring all 2,989,339 released predictions reproduces published median angular errors within 0.055°: 23.6548° versus 23.60° for CC/track and 56.1553° versus 56.12° for NC/cascade. This misses a deliberately strict 0.01° identity tolerance. On 256 development events, checkpoint and released directions differ by 0.944° median (7.708° p95); therefore the clean gate remains open. A module-dropout positive control is monotone (15.28° → 25.12° median angular error; 10-NN retention 1.000 → 0.393 from 0–50% dropout), establishing perturbation, hook, and paired-metric feasibility without presenting the pilot as a scientific claim.

References

[1] R. F. Ørsøe et al. "NuBench: An Open Benchmark for Deep Learning-Based Event Reconstruction in Neutrino Telescopes." JINST 21 (2026). arXiv:2511.13111.

[2] A. Gretton et al. "A Kernel Two-Sample Test." JMLR 13 (2012), 723–773.

[3] D. Lopez-Paz and M. Oquab. "Revisiting Classifier Two-Sample Tests." ICLR (2017).

[4] C. Guo et al. "On Calibration of Modern Neural Networks." ICML (2017).